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Orbit and Galaxy User-Defined Tools for Reproducible, Auditable Computational Biology

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Five published genomics papers, reanalyzed end-to-end from raw data with an AI agent (**Orbit**) orchestrating Galaxy and its **User-Defined Tools**—and an honest account of what matched, what didn't, and why.

Learn about Orbit
galaxyproject.org/agents/orbit · Auditable agentic analysis on Galaxy.

A Abstract

Standard steps—alignment, peak calling, quantification—rarely break genomic reproducibility. What breaks is the manuscript-specific glue between them: custom filtering, integration, and comparison scripts tied to one machine's local software state. Here, we pair **Orbit**, an AI-enabled analysis environment, with Galaxy's **User-Defined Tools (UDTs)** to execute custom logic on shared HPC and cloud infrastructure as containerized jobs with full Galaxy history provenance. Each UDT pins a versioned container rather than an ephemeral notebook kernel and library snapshot. These custom steps remain executable long after publication without succumbing to dependency drift.

We applied this architecture to reproduce five published genomics studies from raw reads. In Chung et al. (2014), which defined a 97-gene SrbA regulon in *Aspergillus fumigatus*, Orbit discovered references and workflows through BRC-Analytics and ran eight custom UDTs for RNA-seq contrasts, ChIP/RNA integration, and concordance scoring. This pipeline uncovered an input-control depth imbalance that distorted peak calling and recovered **89 of 97 published targets outright**—with seven of the remaining eight traced to annotation coordinate drift rather than missing signal. In our reanalysis of Toenhake et al. (2018) in *Plasmodium falciparum*, UDTs handled GTF chromosome renaming, alignment filtering, and peak-to-gene integration to recover **94–96%** of published correlated gene–peak pairs against the deposited peak universe. Routing custom steps—including UDTs written and orchestrated by Orbit—into containerized Galaxy histories turns one-off analysis glue into shareable provenance records. AI-assisted analyses become at least as inspectable, reproducible, and durable as workflows written by hand.

B How closely did five reanalyses match the published results?

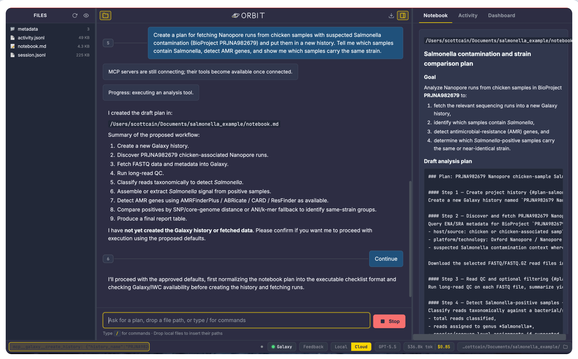
Each bar shows that project's primary match metric—rebuilt from raw public reads on Orbit + Galaxy without copying processed files.



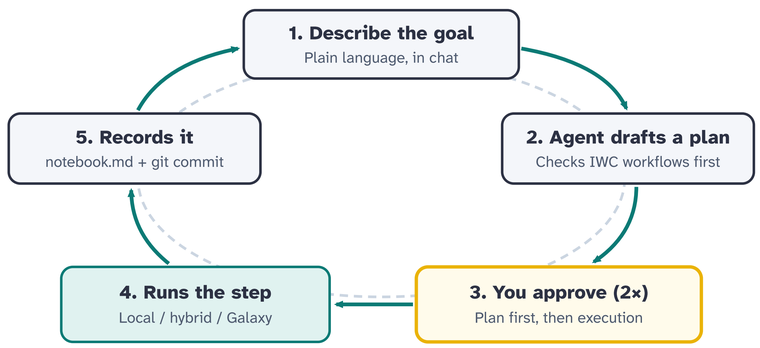
Where a project's own report also flagged a gap (e.g., a 4× peak-count mismatch in Ruiz), the corresponding panel below explains why.

5	67+	3	48+
published papers reanalyzed—5 organisms, 4 assay types	custom Galaxy tools built where none existed	real Galaxy / IWC bugs found and fixed as a side effect	dated git commits forming an auditable project record

C How a session runs



Orbit in an actual session—project files, chat/plan, and executable `notebook.md` panes.



No compute executes, and nothing enters the record, until the researcher approves it—twice.

Tool Source for srba-chip-rna-integrator

```
class: GalaxyUserTool
id: srba-chip-rna-integrator
name: SrbA ChIP RNA integrator
version: 0.1.0
description: Assign Af293 SrbA ChIP peaks to genes and merge 120-minute RNA-seq
    fold-change contrasts.
container: python:3.11-slim
shell_command: |-
  python <<'PY'
  import csv, math, re
  from collections import defaultdict
  from pathlib import Path

  ANNOTATION = Path('$(inputs.annotation.path)')
  PEAKS = Path('$(inputs.peaks.path)')
  CONTRAST_A = Path('$(inputs.dsrba120_all.path)')
  CONTRAST_B = Path('$(inputs.dsrbb120_all.path)')

  PROMOTER_WINDOWS = {'promoter_1kb': (1000, 200), 'promoter_2kb': (2000, 200)}
  NEAR_TSS_CUTOFF = 2000
  KEY_TARGET_SYMBOLS = {'srbA','srbB','erg3','erg24','erg25','erg250','cyp51A','erg11A','cyp51B','erg11B','hem13','niiA','alcC','sit1','hyd1'}
  KEY_TARGET_IDS = {'AFUA_2G01260','AFUA_4G03460','AFUA_4G08090','AFUA_7G03740','AFUA_4G04820','AFUA_8G02440','AFUA_1G07480','AFUA_1G12840','AFUA_5G06240','AFUA_7G

  def parse_gtf_attrs(s):
      d = {}
      for part in s.split(';'):
          part = part.strip()
          if not part or ' ' not in part:
              continue
          k, v = part.split(' ', 1)
          d[k] = v.strip().strip('"')
      return d

  def gene_sort_key(gid):
      m = re.match(r'AFUA_(\d+)G(\d+)', gid or '')
```

Tool Source for srba-chip-rna-integrator

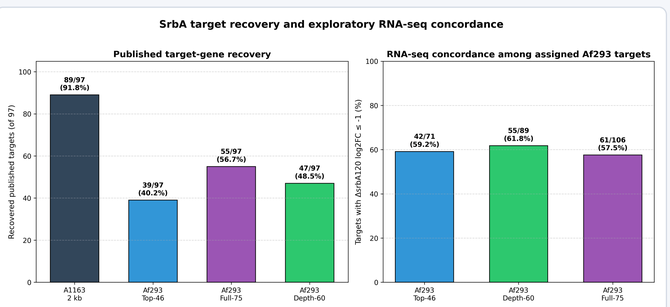
```
outputs:
  - name: overlap_summary
    type: data
    format: tabular
    from_work_dir: key_target_chip_rna_overlap_top46.tsv
  - name: summary
    label: SrbA ChIP RNA integration summary
    type: data
    format: tabular
    from_work_dir: chip_rna_integration_summary.tsv
  - name: report
    label: SrbA ChIP RNA integration report
    type: data
    format: txt
    from_work_dir: chip_rna_integration_report.txt
help:
  format: markdown
  contents: >-
    Assigns Af293 SrbA ChIP-seq MACS2 peaks to nearby genes and joins the
    assignments to 120-minute matched RNA-seq fold-change tables.

    Inputs are an Af293 GTF annotation, a matched-control Af293 MACS2 narrowPeak
    BED, and the Step 9 all-gene tables for WT120 vs dSrbA120 and WT120 vs
    dSrbB120. The tool uses gene-body overlap, promoter windows of 1 kb or 2 kb
    upstream plus 200 bp downstream, and nearest TSS within 2 kb.

    Outputs include all peak-gene assignments, one best peak assignment per
    target gene, targets down in each mutant, canonical/key target overlaps, a
    conservative top-46 peak subset by MACS2 q-value, and a summary/report. Fold
    changes are exploratory n=1 values from the upstream Step 9 tables.
```

Example UDT (`srba-chip-rna-integrator`): assigns SrbA ChIP-seq peaks to genes and merges them with matched RNA-seq fold changes in one containerized Galaxy job rather than a fragile manual join.

D *Aspergillus fumigatus* • ChIP-seq—Chung et al. 2014

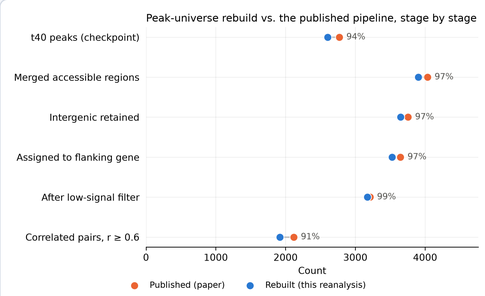


Recovery of 97 published SrbA target genes (left) and RNA-seq concordance among Af293-assigned targets (right) across four peak-calling designs.

- **8 custom Galaxy UDTs** ran the manuscript-specific steps: RNA-seq contrasts, ChIP/RNA integration, and paper-concordance scoring.
- **A depth imbalance between input controls**—easy to miss in a notebook—surfaced as an inspectable Galaxy job and was corrected.
- **89 of 97 published SrbA target genes** recovered outright—seven of the remaining eight traced to a decade of annotation coordinate drift rather than lost signal.

▶ Run → Check → ! Catch → ✓ Fix & record **Caught its own mistake:** a "verified" tool link meant nothing—Orbit re-audited by exporting the actual UDT definitions.

E *Plasmodium falciparum* • ATAC + RNA-seq—Toenhake et al. 2018

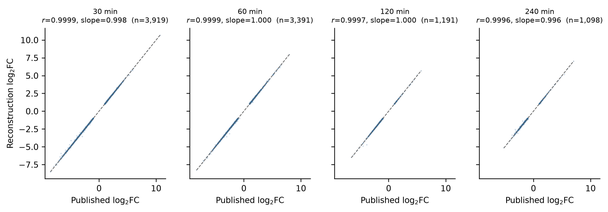


Peak-universe rebuild vs. published pipeline stage by stage—published (orange) vs. reanalysis (blue), with % match at each step.

- **UDTs handled chromosome renaming, alignment filtering, and peak-to-gene integration** across two genome annotation releases, run side by side on purpose.
- **94–96% of correlated gene–peak pairs** recovered from the authors' deposited peaks, and **91% recall / 96% precision** when rebuilding the peak universe from raw reads alone.
- **An apparent 4× shortfall** turned out to be a smoke-test tool mistaken for a peak caller—not a flaw in the published method.

▶ Run → Check → ! Catch → ✓ Fix & record **Declared the paper "irreproducible,"** then found its own placeholder peak-caller—a named, versioned UDT—and reversed the verdict in 24 hours.

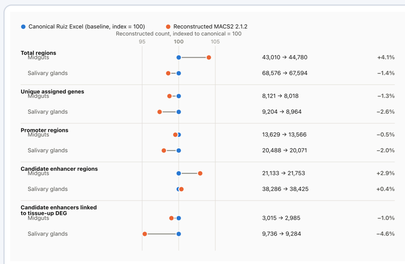
F *A. fumigatus* RNA-seq—Hokken et al. 2018



Reanalysis $\log_2$FC vs. published $\log_2$FC at each timepoint (r and slope shown, dashed line $y = x$).

- **DEG counts initially missed the paper by 31–56%** at later timepoints—traced to a modeling choice, not bad data.
- **Fixed to within 2.4%** of published counts, with retired AspGD tools replaced by custom GO-enrichment UDTs.
- **Checked its own workflow design against IWC** and overturned its earlier assumption when the evidence didn't support it.

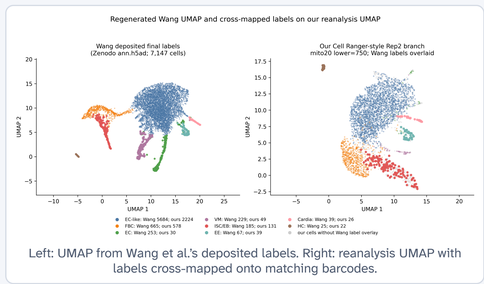
G *Anopheles gambiae* ATAC-seq—Ruiz et al. 2021



Canonical (paper) vs. reconstructed region counts across five regulatory summaries.

- **One UDT failed three different ways** in a single afternoon—each diagnosed, fixed, and logged by version.
- **Peak coordinates reproduced with zero free parameters (92%)**, whereas published peak counts did not—and the report states why plainly.
- **Pre-registered a blind cross-tissue validation test** rather than trust a fitted parameter, and reported "inconclusive" when that was the honest answer.

H *Aedes aegypti* single-cell—Wang et al. 2024



Left: UMAP from Wang et al.'s deposited labels. Right: reanalysis UMAP with labels cross-mapped onto matching barcodes.

- **Filed 3 real upstream Galaxy / IWC bugs** and contributed 3 reusable skills that benefit workflows beyond this study.
- **Recovered silently erased notebook entries** from the raw activity log two months after a background-process bug.
- **Built a Cell Ranger-style reference** for an *Ae. aegypti* assembly with no existing Galaxy recipe and resolved non-standard mitochondrial gene names.

AI-assisted analysis doesn't have to trade away reproducibility. Captured as versioned, auditable Galaxy jobs rather than opaque local scripts, it can be at least as inspectable and durable as analysis written entirely by hand.

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